



March 14, 2025

Mr. Faisal D'Souza, NCO

Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation

On behalf of the Office of Science and Technology Policy Council (OSTP)

2415 Eisenhower Avenue

Alexandria, VA 22314

[REDACTED]

[REDACTED]

Re: Comments on the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear Mr. D'Souza:

This version of our comments supersedes the version submitted previously because the previous version inadvertently omitted the following statement:

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

The American Chemistry Council's (ACC)¹ Regulatory & Scientific Affairs Division appreciates the opportunity to provide the following comments to the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) of the National Science Foundation, which is working on the behalf of the Office of Science and Technology Policy (OSTP).

ACC is a trade association that represents more than 190 companies engaged in the business of chemistry. Our members play a foundational role in U.S. economic production and performance. Our member operations apply

¹ The American Chemistry Council's mission is to advocate for the people, policy, and products of chemistry that make the United States the global leader in innovation and manufacturing. To achieve this, we: Champion science-based policy solutions across all levels of government; Drive continuous performance improvement to protect employees and communities through Responsible Care®; Foster the development of sustainability practices throughout ACC member companies; and Communicate authentically with communities about challenges and solutions for a safer, healthier and more sustainable way of life. Our vision is a world made better by chemistry, where people live happier, healthier, and more prosperous lives, safely and sustainably—for generations to come.



the science of chemistry to make innovative products, technologies, and services. The expertise and efforts of our members will be essential to securing and advancing American AI dominance.

Our comments follow. If you have questions or concerns, please contact me.

Jessica Ryman, PhD, DABT

Senior Director, Chemical Management



Chemical Industry-Specific Comments

1. American chemistry is instrumental for AI dominance

The U.S. has historically been a net exporter of chemicals, contributing positively to the nation's trade balance. **The U.S. chemical industry posted a trade surplus of \$30.1 billion in 2023.** With the exception of Europe, the U.S. chemical industry held a trade surplus with every other major region of the world, including Asia.

Chemistry makes the U.S. economy possible. Chemistry products support nearly every corner of the economy. In fact, **a quarter of U.S. GDP is generated from industries that rely on resilient supply chains for chemistry**, including high-tech goods, aerospace, automotive goods, healthcare, agriculture and food production, among others. More than 80% of basic & specialty chemicals are purchased directly by the industrial sector. Chemistry is consumed by the manufacturing sector, both directly and indirectly.

1.1 Hardware and chips

Quite simply, developing and manufacturing electronics would not be possible without chemistry. Making everything from semiconductors to printed circuit boards to display materials, and many other electronic components requires a multitude of chemicals. In fact, \$1 for every \$10 worth of materials used to produce electronic components is spent on chemistry, and a single semiconductor chip requires 500 different chemistries.

1.2 Data centers

In addition to the products we use every day such as mobile phones, laptops, and electric vehicles, electronics and the chemistry that goes into them are becoming increasingly more important to building data centers and artificial intelligence (AI). For example, chemistry is important for the building of data centers and the systems that control them (i.e., fire suppressant systems, HVAC systems, water cooling etc.).

1.3 Energy consumption and efficiency²

Our Official Policy Position on Energy is that ACC supports a national energy strategy that develops and uses America's diverse sources, promotes energy efficiency, and encourages the development and adoption of new technologies.

² Learn more at: <https://www.americanchemistry.com/better-policy-regulation/energy>

Chemistry is the science behind innovative solutions that empower diverse energy options, create jobs, and propel U.S. economic growth. Chemistry enables energy-saving and renewable applications including solar panels and wind turbines, electric and fuel-efficient vehicles, high-performance building materials, advanced batteries, energy-efficient lighting, and more.

2.0 American chemistry for AI needs smart chemical regulation

2.1 More hardware and chips made in America means more American chemistry

More chemistry will be needed to supply the return of electronics manufacturing to the U.S. To meet this growing demand and keep chemical manufacturing in America, policymakers must support the production of the chemistries needed to develop new technologies and make new electronics.

2.2 Smart and timely regulation for development and manufacture of vital chemistries is needed

Congress understood that protecting human health and the environment and promoting economic and societal advancement are not mutually exclusive when it amended the Toxic Substances Control Act (TSCA) in 2016. In fact, sound chemical management policies are critical to American innovation and competitiveness. However, TSCA is not working as intended by Congress and the Environmental Protection Agency (EPA) is not meeting its statutory mandates.

As the demand for innovative chemistries grows, the backlog of new chemicals awaiting TSCA review beyond the 90-day period is impeding investments in the production of new and innovative chemicals here in the United States. Seventy percent of ACC member companies surveyed reported choosing to introduce new chemicals outside the U.S. due to uncertainties and challenges with the EPA's New Chemicals Program³⁴.

³ [Chris Jahn testimony on the Impact of TSCA and the Lautenberg Chemical Safety for the 21st Century Act before the House Energy and Commerce Environ... - American Chemistry Council](#)

⁴ Learn more at: <https://www.americanchemistry.com/better-policy-regulation/chemical-management/toxic-substances-control-act-tsca/tsca-new-chemicals-under-review-tracking>

ACC Comments Specific to Requested Comments

3.1 Model development would (ideally) involve subject-matter experts

The involvement of subject-matter experts in the development of AI models for use in specific areas of expertise would (ideally) involve subject-matter experts.

3.2 Open source development

ACC notes the potential for greater security and bias concerns with open source development because the development environment is less controlled. As such, it may be advisable to limit open source development to lower-risk applications. Furthermore, open source AIs would likely need additional review prior to deployment.

3.3 Application and use: private sector and government

ACC notes that AI has high potential for multiple applications and uses in both the private sector (to include the chemical industry) and government (to include chemical regulation) that could increase efficiency. Assuming that governments would generally be expected to have more stakeholders than individual businesses, the use of AI in government would generally be expected to be higher risk. (That said, the potential for individual companies that specialize in AI to amass a considerable amount of global power cannot be ruled out). For government in particular, if models are developed, inclusion of subject-matter experts is highly recommended, as is starting out on smaller, lower-risk projects prior to expansion into higher-risk areas.

3.4 Explainability and assurance of AI model outputs

AI models should be of sufficient explainability and AI model inputs and outputs should be sufficiently transparent to facilitate external validation and reproducibility when used for government activities that allow for public comment opportunities. Furthermore, such AI models should be transparent about what policies and assumptions are applied and how these affect the outcome. (The TREAT framework of Trustworthy, Robust, Ethical, Accountable, and Transparent AI may be helpful in this contexts and other contexts in relation to AI). We further underscore the necessity for human oversight for quality assurance, during both AI development and for model outputs, to identify and avoid issues such as drift, AI ‘hallucinations’, bias, and other undesirable outcomes.

3.5 Cybersecurity

Any use of AI by federal regulatory agencies that requires the submission of proprietary data or confidential business information should ensure appropriate cybersecurity measures in accordance with the agencies’ Congressional authorities. We further note that cybersecurity also comes into play during

AI development to prevent ‘hallucinations’. Human oversight will be instrumental in ensuring cybersecurity.

3.6 Data privacy and security throughout the lifecycle of AI system development and deployment (to include security against AI model attacks)

Any use of AI by federal regulatory agencies that requires the submission of proprietary data or confidential business information should be conducted in accordance with Congressional authority and ensure data privacy and security not only during AI system development and deployment, but also AI system updates and AI system decommissioning. A federal regulatory agency should also provide notice to the regulated community that its proprietary data or confidential business information may be processed or analyzed using AI.

3.7 Risks

AI comes with many potential risks that should be assessed and managed. These include, but may not be limited to: loss of safety (e.g., AI could cause danger or damage to people, property, or the environment); loss of security and resiliency (e.g., loss of integrity, confidentiality, and loss of ability to function in the face of change); instability, unreliability, and invalidity, (e.g., AI could be inaccurate and not robust); AI could become unexplainable and uninterpretable; AI could become unaccountable and untransparent; AI could be unfair and biased; and AI could result in loss of privacy.

3.8 Regulation and governance

As far as we are aware, there is little regulation and governance of AI in the United States. We note that, in general, whatever governance structure(s) is/are developed and applied, these should ensure that AI has human oversight by trained humans that ideally includes a risk governance structure (i.e., risk mapping, risk measurement, and risk management). Risks should be appropriately assessed and mitigated such that AI (in general) will be safe (e.g., not endanger people, property, or the environment), secure and resilient (e.g. maintain integrity, confidentiality, and availability in the face of change), stable and reliable, valid (e.g., accurate and robust), explainable and interpretable, accountable and transparent, fair and unbiased, and maintaining of privacy.

3.9 Technical and safety standards

No comment.

3.10 National security and defense

No comment.

3.11 Research and development (R&D)

R&D can encompass two areas: R&D for AI itself and use of AI in R&D to advance discovery and innovation

3.11.1 R&D for AI itself

R&D for AI should include enough users during testing to ensure that unanticipated biases and other risks are identified and mitigated early on. Additionally, R&D should include subject-matter experts for AIs that are to be developed and deployed in specific areas of subject matter expertise.

All AI models should be validated with test data that differs from the training data. Additionally, consequential models should undergo peer review, to include peer-review of the model output. The output of the model could be compared (by humans) to reference output, including by appropriate statistical methods (such as linear regression, when statistically relevant). Additionally, for consequential models, the references and data utilized should be available for external validation. We further note that comparison of model output over time, as well as maintaining the initial data used to develop the model, should also help with identifying any model drift over time.

3.11.2 Using AI to advance discovery and innovation

We note that the AI models have accelerated R&D in the pharmaceutical industry and we note the high potential for AI to advance discovery and innovation in other industries.

3.12 Education and workforce

Unbiased or leave blank.

3.13 Innovation and competition

We note that AI will likely be helpful for innovation, but that it would be preferable for AI to be used as an assistive tool to aid users in innovating.

3.14 Intellectual property

We have noted previously that AI used by federal agencies should protect industry intellectual property (e.g. when regulatory agencies require the submission of proprietary data or confidential business information).

3.15 Procurement

No comment.

3.16 International collaboration

We anticipate that AI will be integrated over time into currently existing international collaborations (e.g. OECD and other forums). We further note that convening international expert groups on various topics in international collaboration provides an opportunity for SME input in the development of AI that will be used internationally.

3.17 Export controls

No comment.